# COMP438/538 Reinforcement Learning

## Model-Free Prediction

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# Models and Prediction without the Models

So far:

- MDPs
  - State, Action, Reward, Transition Dynamics, Discount
  - Optionally terminal and start states
- Returns, Values and Q-Values

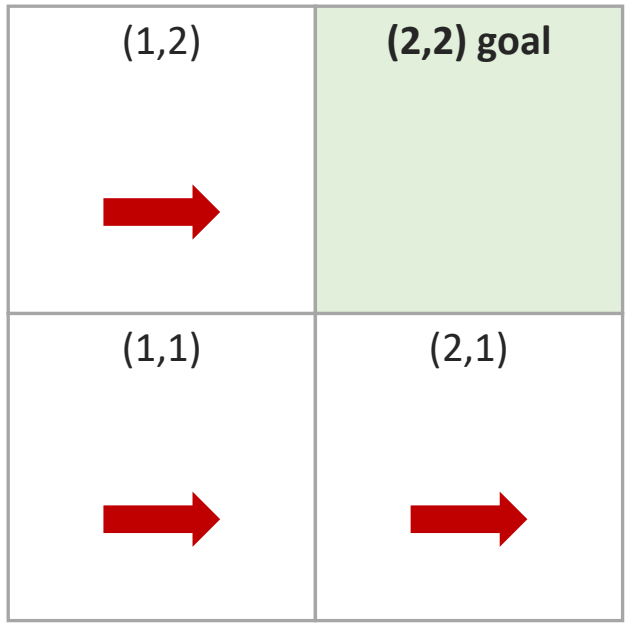
- Policy Evaluation
- Policy Iteration (Evaluation then Selection)
- Value Iteration (Concurrent Evaluation and Selection)

- Which computations need the transition and reward model?
- Which action-selection step can use a learned table directly?

# Policy evaluation without the model

- With a model: evaluate a policy, then improve it.
- Today: evaluate the **same fixed policy** using observed experience.
- Next week: improve policies using experience.

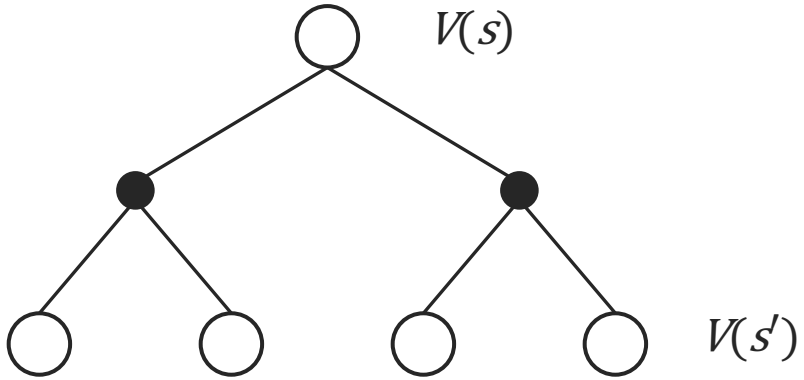
**Question:** You can run this policy and observe what happens. How could you estimate $v^{\pi}((1,1))$?

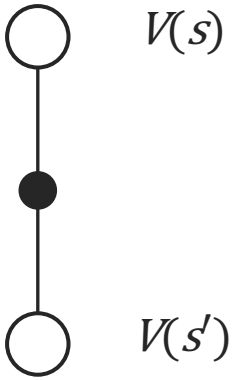



**Always choose right; $\gamma$=0.5.**

Transition probabilities: **?**

# Why learn from samples?




| Expected backup | Sample backup |
|---|---|
| **Average over possible outcomes using the model** | Use an observed transition |
| **Need outcome probabilities and expected rewards** | Receive $(S_t, A_t, R_{t+1}, S_{t+1})$ |

- A robot's dynamics may be unknown or unreliable.
- A simulator may produce samples even when enumerating all outcomes is impractical.
- One cheap update can still require many samples to learn accurately.

Reinforcement Learning: Models are unknown ahead of time (Today: Sampling, Later: Model-Learning)

# Side Note: Three different questions about the data

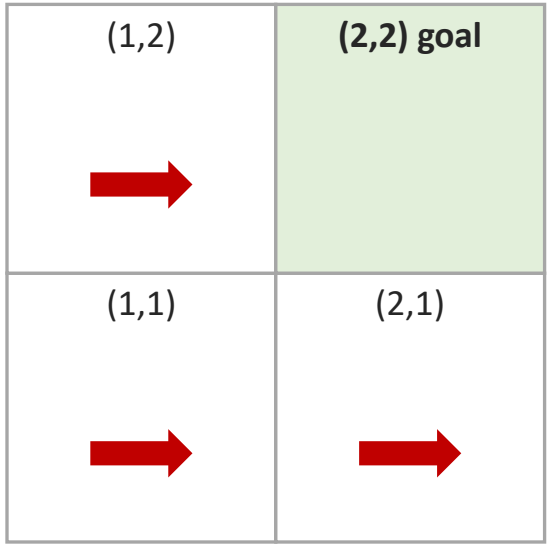
| Question | Distinction |
| --- | --- |
| **Do we use or learn an environment model?** | Model-based / model-free |
| **Can we collect more interaction data while learning?** | Online / offline (fixed data) |
| **Does the behaviour generating data match the policy being evaluated?** | On-policy / off-policy |

**Today: tabular, model-free, on-policy prediction of a fixed $\pi$.**

# One observed episode

**(1,1) → (1,1) → (1,2) → goal**

| $t$ | $S_t$ | $A_t$ | $R_{t+1}$ | $S_{t+1}$ |
|---|---|---|---|---|
| **0** | (1,1) | right | −0.04 | (1,1) |
| **1** | (1,1) | right | −0.04 | (1,2) |
| **2** | (1,2) | right | +1 | goal |




Important concepts:

- Transitions
- Trajectories/Roll-outs
- Episode
- Return
- Averaging Returns

# Monte Carlo: value as a mean return

$$v^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$$

**Monte Carlo prediction:** follow $\pi$, observe complete episodes, and average returns after visits to each state.

$$G_T = 0, \qquad G_t = R_{t+1} + \gamma G_{t+1}$$

Episode: (1,1) → (1,1) → (1,2) → goal; rewards −0.04, −0.04, +1; $\gamma = 0.5$.

**Your turn:** calculate $G_2$, then $G_1$, then $G_0$ for our episode.

$$G_2 = 1, \qquad G_1 = -0.04 + 0.5(1) = 0.46, \qquad G_0 = -0.04 + 0.5(0.46) = 0.19$$

**One observed return is a sample, not the expectation.**

# Which visits count?

| Visit to (1,1) | Return following that visit |
|---|---|
| **First, $t$=0** | 0.19 |
| **Second, $t$=1** | 0.46 |

Episode: (1,1) → (1,1) → (1,2) → goal

- **First-visit MC:** include only the first visit to a state in each episode.
- **Every-visit MC:** include every visit to that state.

**Question:** what sample mean does each produce after this one episode? (For (1,1))

**Answer:** first-visit $0.19$; every-visit $(0.19 + 0.46)/2 = 0.325$.

# From an average to an incremental update

For one newly included return $G_t$, first increment $N(s)$:

$$N(s) \leftarrow N(s) + 1$$

$$V(s) \leftarrow \frac{(N(s) - 1)V(s) + G_t}{N(s)} = V(s) + \frac{1}{N(s)}[G_t - V(s)]$$

**A common update form**

$$V(S_t) \leftarrow V(S_t) + \alpha[U_t - V(S_t)]$$

$U_t$: target. $\alpha$: fraction of the discrepancy used for this update.

| Step size for MC | Meaning |
|---|---|
| $1/N(s)^*$ | Exact sample mean of included returns |
| **Constant $\alpha$** | Recency weighting; initialization can still matter, better for non-stationary problems |

*This is a special case of decayed $\alpha$ that meets the Robbins–Monro conditions for convergence, but not all decay schedules work

# First-visit MC prediction

1. Fix $\pi$. Initialize $V(s) = 0$ and $N(s) = 0$ for nonterminal states.
2. Generate a complete episode following $\pi$.
3. Compute its returns backwards: $G_T = 0$, then $G_t = R_{t+1} + \gamma G_{t+1}$.
4. Scan the episode in time order. At the **first visit** to each state:
   - increment $N(S_t)$;
   - update $V(S_t) \leftarrow V(S_t) + \frac{1}{N(S_t)}[G_t - V(S_t)]$.
5. Repeat with another episode.

**For constant-step MC, replace $1/N(S_t)$ by $\alpha$ in step 4. The first-visit rule stays.**

**Question:** How do the steps change if we do the every-visit version?

# Driving home: must learning wait?

## Predict the minutes remaining until home

| Situation | Elapsed minutes | Predicted minutes remaining | Predicted total minutes |
|---|---|---|---|
| **Leave office** | 0 | 30 | 30 |
| **Reach car; raining** | 5 | 35 | 40 |
| **Exit highway** | 20 | 15 | 35 |
| **Behind slow truck** | 30 | 10 | 40 |
| **Enter home street** | 40 | 3 | 43 |
| **Arrive home** | 43 | 0 | 43 |

## At the car, would you revise the office prediction now, or wait until home?

## Another Question: What happens if the problem is continuing?

Standalone prediction example: rewards are elapsed minutes; $\gamma = 1$. Book Example 6.1 (Sutton & Barto, 2018).

# A sampled Bellman target – Temporal Difference (TD) Learning

**1.** Recall the Bellman expectation equation:

$$v^{\pi}(s) = \mathbb{E}_{\pi}[R_{t+1} + \gamma v^{\pi}(S_{t+1}) \mid S_t = s]$$

**2.** Use one observed transition and the current next-state estimate:

$$U_t = R_{t+1} + \gamma V(S_{t+1})$$

**3.** Compare the target with the current estimate, then update:

$$\delta_t = U_t - V(S_t), \qquad V(S_t) \leftarrow V(S_t) + \alpha\delta_t$$

**Bootstrapping:** use an existing value estimate in a new update target.

The expected TD target is a Bellman backup of the current $V$; it need not equal $v^{\pi}$.

# TD(0) prediction

Fix $\pi$ and choose $\alpha$. Initialize the nonterminal estimates; set $V(\text{terminal}) = 0$.

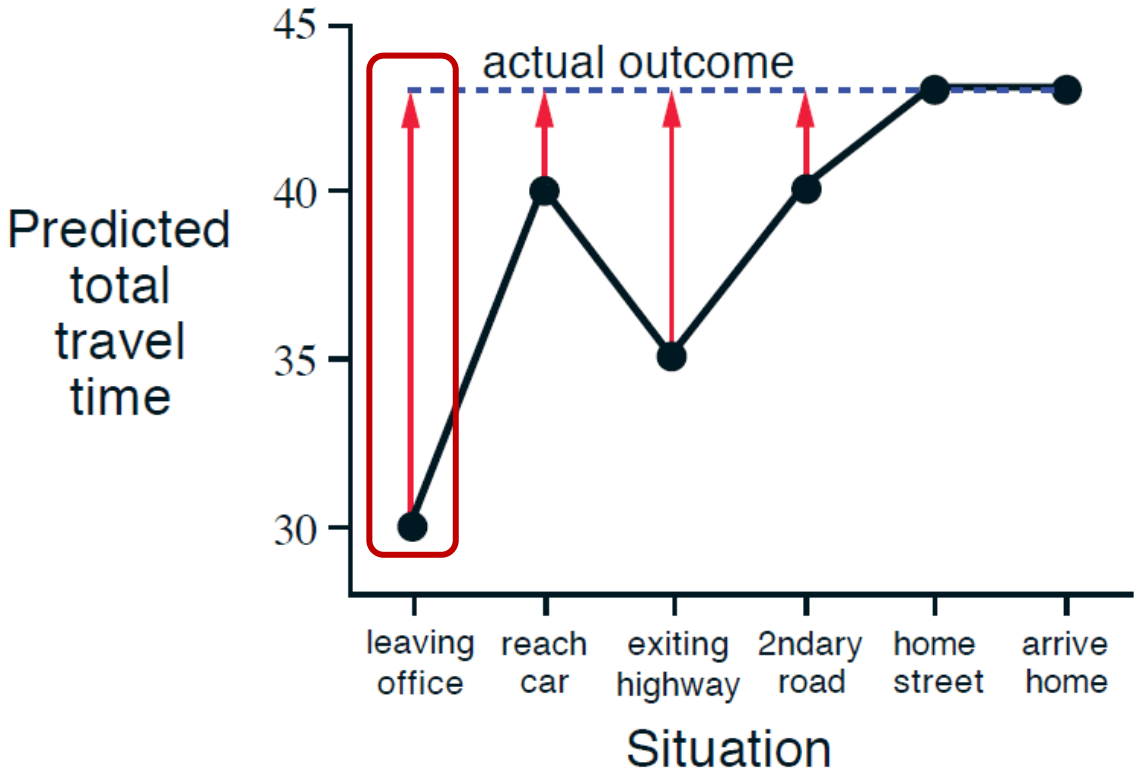
At each transition:

1. Take $A_t$ according to $\pi$; observe $R_{t+1}$ and $S_{t+1}$.
2. Form $U_t = R_{t+1} + \gamma V(S_{t+1})$.
3. Update $V(S_t) \leftarrow V(S_t) + \alpha[U_t - V(S_t)]$.
4. Advance to the next state; if terminal, end the episode.

**Update after observing the transition. Read the current table before changing it.**

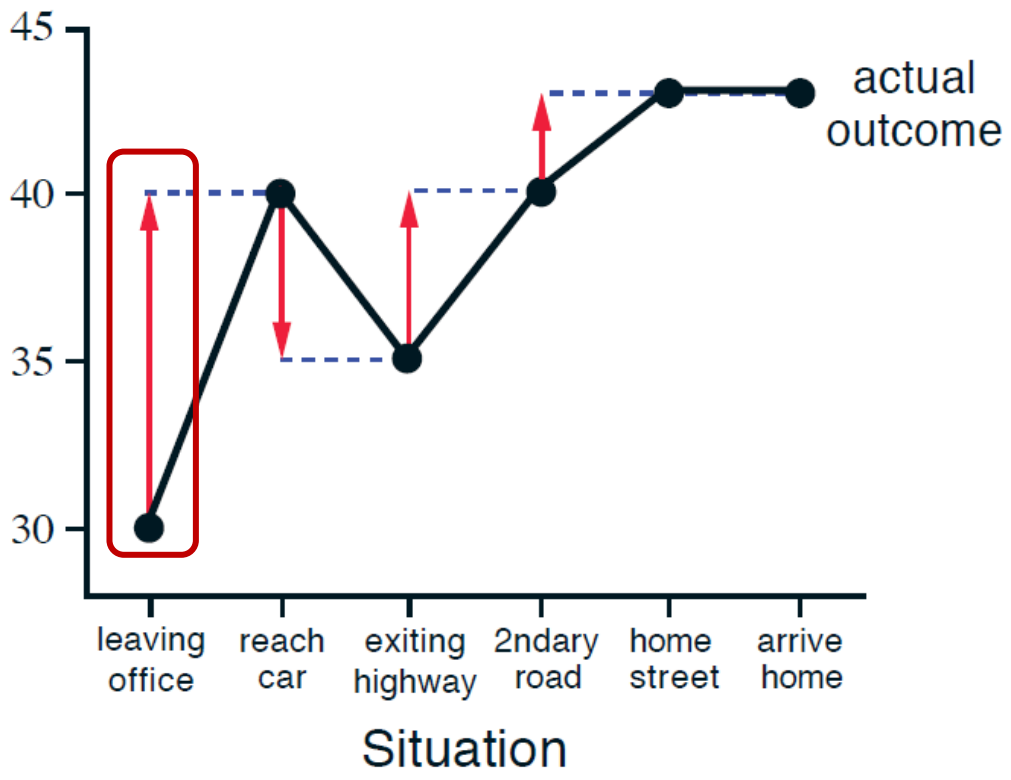
# Driving home: what each update changes

**MC: toward the final outcome**




First arrow: office prediction $30 \to 43$, after reaching home

**TD: toward the next prediction**




First arrow: office prediction $30 \to 5+35=40$, after reaching the car

$\gamma=1$; $\alpha=1$ for the full arrows. Figure: Sutton & Barto (2018), Figure 6.1.

# Your turn: MC and TD on the same episode

(1,1) → (1,1) → (1,2) → goal;  rewards −0.04, −0.04, +1;  returns 0.19, 0.46, 1

**Reset each method independently:** $V = 0$, $\gamma = 0.5$, **constant** $\alpha = 0.5$.

1. First-visit MC: update each state's first visit using its completed return.
2. TD(0): update after each transition, in time order.
3. Why does the goal reward affect the estimates differently?

| Method / event | Target $U_t$ | Estimate being changed | Value after update |
|---|---|---|---|
| **MC, first (1,1) visit** | 0.19 | $V((1,1))$ | 0.095 |
| **MC, first (1,2) visit** | 1 | $V((1,2))$ | 0.5 |
| **TD, $t$=0** | −0.04 | $V((1,1))$ | −0.02 |
| **TD, $t$=1** | −0.04 | $V((1,1))$ | −0.03 |
| **TD, $t$=2** | 1 | $V((1,2))$ | 0.5 |

MC and TD have different targets and update schedules; this is not a performance ranking.

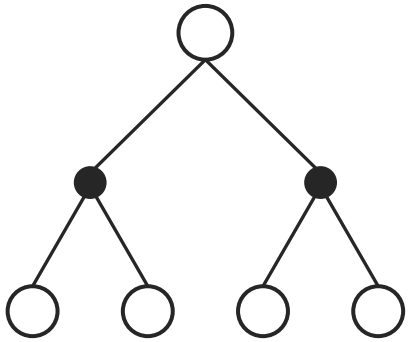
# One episode versus an expected value

| State | First-visit MC after this episode | TD(0) after this episode | True $v^{\pi}$ from the known model |
|---|---|---|---|
| **(1,1)** | 0.095 | −0.03 | ≈ 0.051 |
| **(1,2)** | 0.5 | 0.5 | ≈ 0.836 |

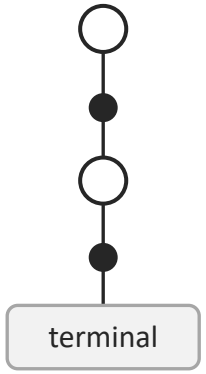
**The right-hand column is a checking benchmark; the learner did not use it.**

**Questions:** Can this episode establish which method is better? What would we need to compare learning after many episodes?

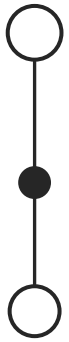
# Sampling and bootstrapping



DP: all successors, one step




MC: one path to termination



TD(0): one sampled step

| Method | Uses sampled experience? | Bootstraps from value estimates? | What supplies the target? |
|---|---|---|---|
| **DP expected evaluation backup** | No | Yes | Model-weighted reward and successor values |
| **Complete-return MC** | Yes | No | Observed return through termination |
| **TD(0)** | Yes | Yes | One observed reward and next-state estimate |

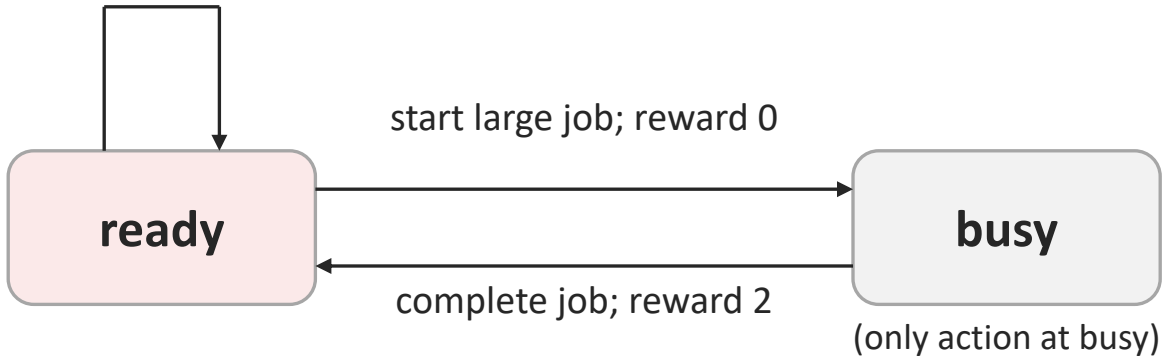
**Sampling:** use observed outcomes in place of a full expectation. **Bootstrapping:** use a value estimate in the target.

**Check:** Which needs a model? Which must wait for the completed return? Which can update before the episode ends?

# A task that keeps running

## A continuing service system — equal-duration time steps

steady service; reward 0.8




Starting from ready:

| Fixed policy at ready | Reward sequence |
|---|---|
| **Always steady service** | 0.8, 0.8, 0.8, … |
| **Always start a large job** | 0, 2, 0, 2, … |

**There is no terminal state. What should good long-run performance mean?**

# Discounted return and average reward

**Average reward: expected reward per step over a long run**

For the policies in this service example, starting from ready:

$$r(\pi) \doteq \lim_{t\to\infty} \frac{1}{t} \mathbb{E}_\pi[R_1 + R_2 + \cdots + R_t \mid S_0 = \text{ready}]$$

| Policy | Average reward per step | Discounted return from ready, $\gamma$=0.5 |
|---|---|---|
| **Steady service** | 0.8 | 1.6 |
| **Large jobs** | (0 + 2)/2 = 1 | 4/3 ≈ 1.333 |

**The objective changes the ranking:** discounted return prefers steady service; average reward prefers large jobs.

Self-Study Suggestion: Sutton & Barto §10.3 for the differential return and values that average-reward methods learn.

# What can we learn before the episode ends?

- MC updates toward a completed observed return.
- TD updates toward an observed reward plus estimated continuation.
- Both can evaluate a fixed policy from experience.

Which needs the next state's estimate? Which needs the completed return?

**Next session:** bias, variance, n-step targets and evaluating learning.

**Next topic:** improve policies using learned action values.

# Can one episode decide?

We evaluated a fixed policy from experience:

$$V(S_t) \leftarrow V(S_t) + \alpha[U_t - V(S_t)]$$

$$U_t = G_t \quad (\text{MC}), \qquad U_t = R_{t+1} + \gamma V(S_{t+1}) \quad (\text{TD}(0))$$

Our grid episode gave different value updates.

(Side note $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$is called the TD-Error)

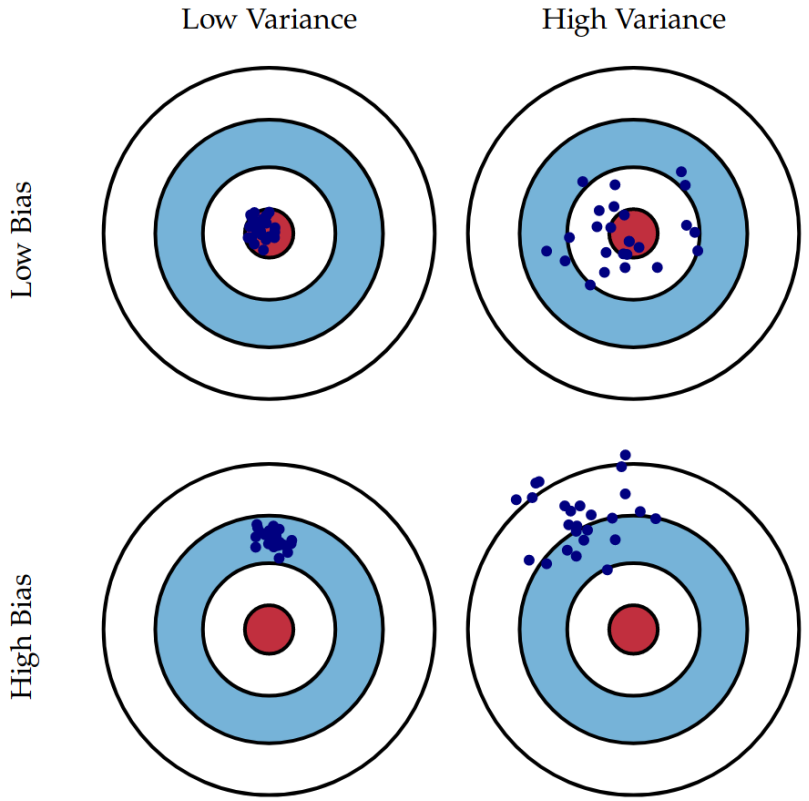
**Question:** Does one episode tell us which method learns better?

**Today:** noisy targets, n-step prediction and evidence for a comparison.

# Bias and variance describe repeated samples

At the same state, repeat the sampling experiment while holding the current table fixed.

- **Bias:** mean target minus the true value.
- **Variance:** spread of targets around their own mean.
- **Mean squared error = variance + bias².**




Illustration, true value 1.0:

| Equally likely targets | Mean | Bias | Variance |
|---|---|---|---|
| **0.0 or 2.0** | 1.0 | 0.0 | 1.0 |
| **0.8 or 1.0** | 0.9 | −0.1 | 0.01 |

# What randomness enters each target?

| | MC | TD(0) |
|---|---|---|
| **Target** | $G_t$ | $R_{t+1}+\gamma V(S_{t+1})$ |
| **Samples** | Remaining episode | One transition |
| **Uses the current value table in the target?** | No | Yes |

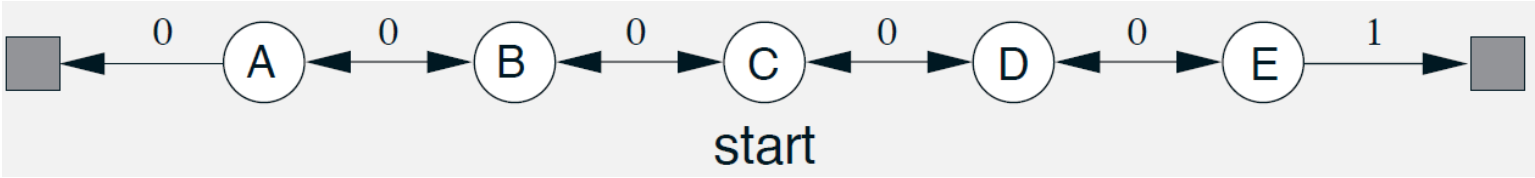
$$\mathbb{E}_\pi[G_t \mid S_t = s] = v^\pi(s)$$

$$\mathbb{E}_\pi[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s] - v^\pi(s) = \gamma\, \mathbb{E}_\pi[V(S_{t+1}) - v^\pi(S_{t+1}) \mid S_t = s]$$

MC has an unbiased **target**. TD's target can inherit error from the table.

**Question:** On the grid, which random transitions affect each target for the start state?

# Example 6.2: random-walk prediction




Five nonterminal states **A, B, C, D, E** between left and right terminals. Start at **C**. Each step moves left or right with probability 0.5.

- **New example:** $\gamma = 1$, +1 on right-terminal entry, 0 on every other transition.
- Both terminals have zero continuation value.
- Start with $V(s) = 0.5$ for each nonterminal state.

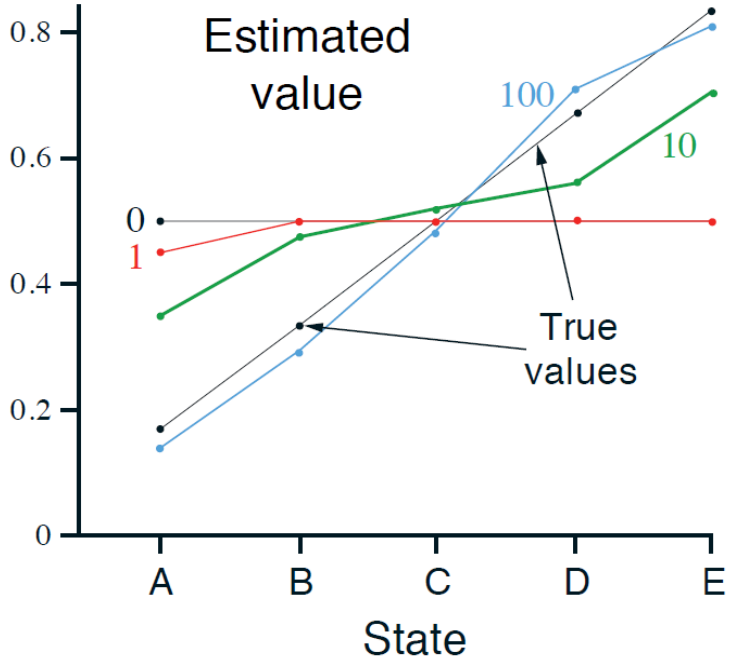
| State | A | B | C | D | E |
|---|---|---|---|---|---|
| **True value** | 1/6 | 2/6 | 3/6 | 4/6 | 5/6 |

**Question:** Why is $v^{\pi}(\mathrm{C}) = 0.5$?

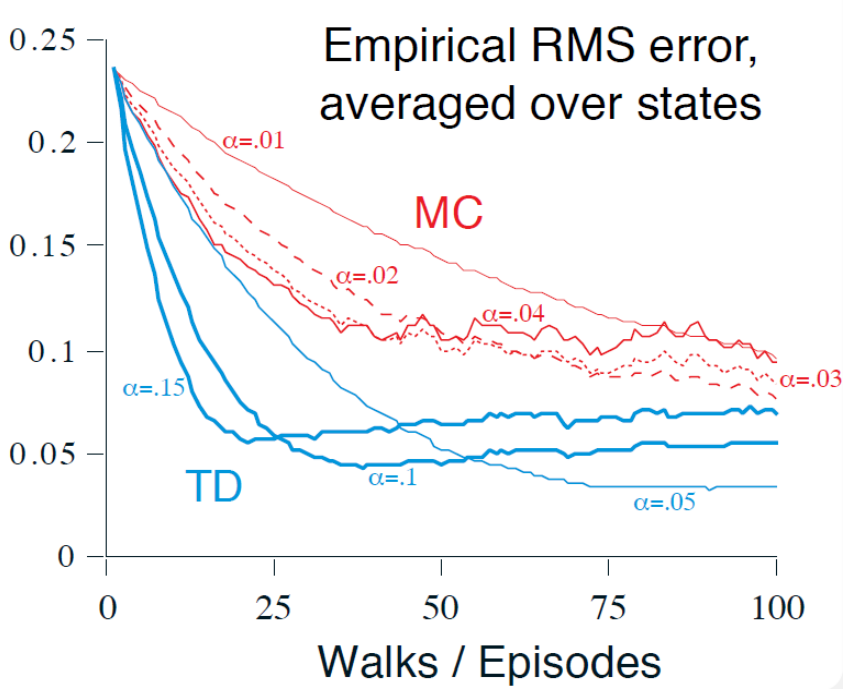
Book Example 6.2 (Sutton & Barto, 2018, p. 125).

# What the random-walk curves show

**Left:** one TD(0) run, $\alpha$=0.1, estimated values by state




**Right:** mean RMS error over 100 runs, several $\alpha$ values, by training episode




**Question:** After the first episode, only V(A) changed. Where did the episode end, and by how much did V(A) change?

**Answer:** left exit. V(A): $0.5 + 0.1(0 - 0.5) = 0.45$

Plots: book Example 6.2 (p. 125). Not Figure 6.2, which shows batch training.

# Learning repeatedly from a fixed batch

We have a fixed set of complete episodes. **No new experience arrives.**

For each pass through the batch:

1. Hold V fixed while computing each visit's increment.
2. Add the increments for each state.
3. Apply those sums, then start the next pass.

$$\text{increment} = \alpha[U_t - V(S_t)]$$

MC uses the observed return. TD uses the reward and the next state's estimate.

**Question:** Can repeated learning from the same data lead to different answers?

# Example 6.4: you are the predictor

**New example:** states A and B, $\gamma = 1$, zero terminal continuation.

Eight observed episodes:

| Count | Episode, with rewards on transitions |
|---|---|
| **1** | A →(0) B →(0) terminal |
| **6** | B →(1) terminal |
| **1** | B →(0) terminal |

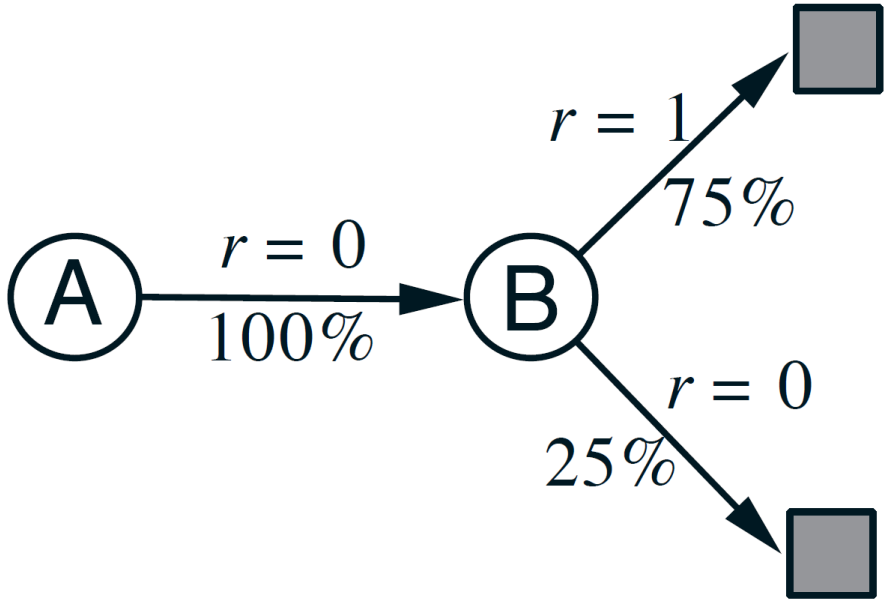
**Question:** What should V(A) and V(B) be after repeated batch learning?

Vote for V(A): **0**, **0.75**, or **depends on the prediction rule**.

Book Example 6.4 (Sutton & Barto, 2018, p. 127).

# Two batch answers from the same data

| | Batch MC | Batch TD(0) |
|---|---|---|
| **B** | $V(\mathrm{B})=6/8=0.75$ | $V(\mathrm{B})=6/8=0.75$ |
| **A** | Its only observed return was 0 | Its observed successor was B, with reward 0 |
| **Answer at A** | $V(\mathrm{A})=0$ | $V(\mathrm{A})=0+V(\mathrm{B})=0.75$ |




Empirical model from the eight episodes

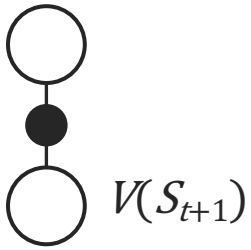
- **MC:** fit the observed returns at each state.
- **TD:** use the observed transitions and share the continuation estimate at B.

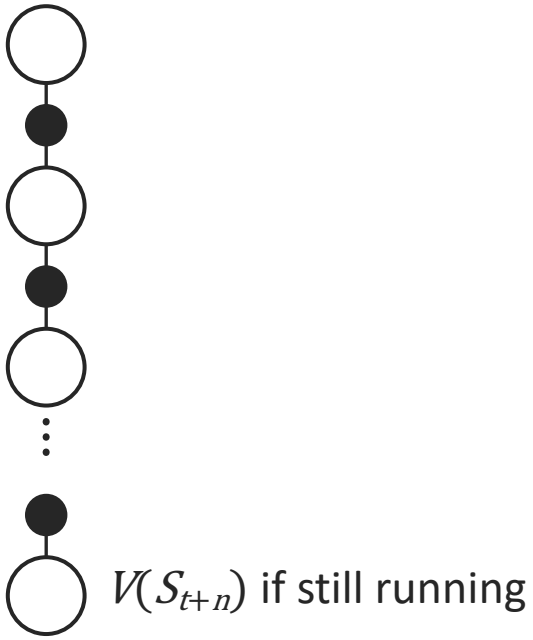
Data: A →(0) B →(0) terminal ×1;  B →(1) terminal ×6;  B →(0) terminal ×1.

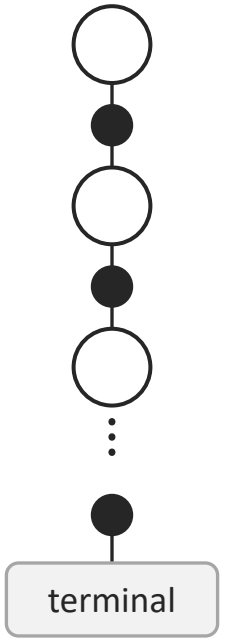
**Question:** When is it reasonable to reuse all the B outcomes after arriving from A?

Book Example 6.4. Both terminal branches have continuation zero.

# Between one transition and a complete return




**TD(0):** one reward, then V at the next state

**n-step TD:** n rewards, then V if the episode continues

**MC:** rewards through termination

**Question:** How much experience should we use before trusting the remaining value estimate?

Sampled backup paths after Sutton & Barto (2018), Figure 7.1. Open circles: states; solid dots: actions.

# The n-step target

Before termination, replace the unobserved tail by the current estimate:

$$G_{t:t+1} = R_{t+1} + \gamma V(S_{t+1})$$

$$G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})$$

For a positive integer n and a visit $t < T$:

If $t + n < T$:

$$G_{t:t+n} = \sum_{i=1}^{n} \gamma^{i-1} R_{t+i} + \gamma^n V(S_{t+n})$$

If $t + n \geq T$:

$$G_{t:t+n} = \sum_{i=1}^{T-t} \gamma^{i-1} R_{t+i} = G_t$$

Same update form when the target becomes available:

$$V(S_t) \leftarrow V(S_t) + \alpha[G_{t:t+n} - V(S_t)]$$

# When can the update happen?

For the visit at time t, wait until **n further transitions or termination, whichever comes first**.

For n = 2:

| Experience observed | Earliest eligible work |
|---|---|
| **One transition** | Wait, unless it terminated |
| **Two transitions** | Update the visit at time 0 |
| **Later transitions** | Update visits as their targets become available |
| **Episode terminates** | Complete all remaining visits in their original order |

Terminal completion needs **no further environment transitions**.

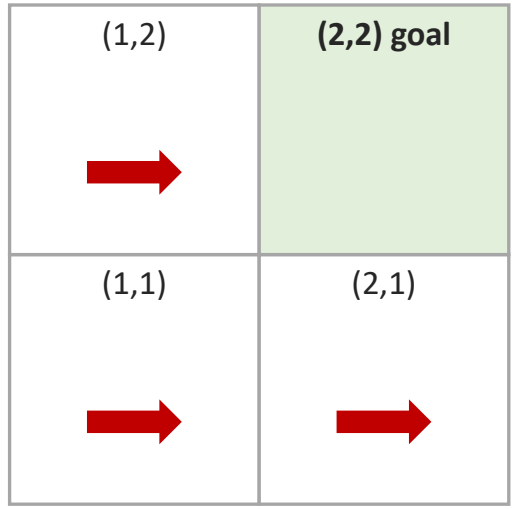
$$V(S_t) \leftarrow V(S_t) + \alpha[G_{t:t+n} - V(S_t)]$$

# Your turn: targets on the grid

Always choose right. $\gamma = 0.5$, $\alpha = 0.5$. Observe:

(1,1) —(−0.04)→ (1,1) —(−0.04)→ (1,2) —(+1)→ goal, $T = 3$

Use this **frozen starting snapshot**, the table after 3.1's TD pass:




| State | (1,1) | (1,2) | (2,1) | goal |
|---|---|---|---|---|
| **V** | −0.03 | 0.5 | 0 | 0 |

1. At $t = 0$, calculate the targets for n = 1, 2, 3 and 5.

2. Using n = 2, make **one isolated update** to V((1,1)).

Hold every table entry fixed while calculating part 1. Part 2 starts from the same snapshot.

# Grid targets and one isolated update

Episode (1,1) →(−0.04) (1,1) →(−0.04) (1,2) →(+1) goal. Snapshot V((1,1)) = −0.03, V((1,2)) = 0.5. $\gamma = \alpha = 0.5$.

| n | Target for the time-0 visit |
|---|---|
| **1** | −0.04 + 0.5(−0.03) = −0.055 |
| **2** | −0.04 + 0.5(−0.04) + $0.5^2$(0.5) = 0.065 |
| **3** | −0.04 + 0.5(−0.04) + $0.5^2$(1) = 0.19 |
| **5** | 0.19 because the episode already ended |

One isolated n = 2 update:

$$V((1,1)) \leftarrow -0.03 + 0.5[0.065 - (-0.03)] = 0.0175$$

**Question:** For n = 5, why is there no extra discounting after the goal?

# Reset, then update in time order

**Reset to V((1,1)) = −0.03, V((1,2)) = 0.5. Use n = 2 and $\gamma = \alpha = 0.5$**

(1,1) →(−0.04) (1,1) →(−0.04) (1,2) →(+1)

| When | Visit updated | Target | New estimate |
|---|---|---|---|
| **After transition 1** | None | — | No change |
| **After transition 2** | $t$=0, (1,1) | 0.065 | 0.0175 |
| **At termination, after transition 3** | $t$=1, (1,1) | −0.04 + 0.5(1) = 0.46 | 0.23875 |
| **Finish pending work, no new transition** | $t$=2, (1,2) | 1 | 0.75 |

$$0.0175 + 0.5(0.46 - 0.0175) = 0.23875, \qquad 0.5 + 0.5(1 - 0.5) = 0.75$$

Same trajectory, delayed information, multiple visits.

# A larger random walk to test the choice of n

| | Earlier five-state walk | New 19-state walk |
|---|---|---|
| **Start** | C | Middle state |
| **Moves** | Left/right with probability 0.5 | Same |
| **Terminal-entry rewards** | Left 0, right +1 | **Left −1, right +1** |
| **Other rewards / γ** | 0 / 1 | 0 / 1 |
| **Initial nonterminal V** | 0.5 | **0** |

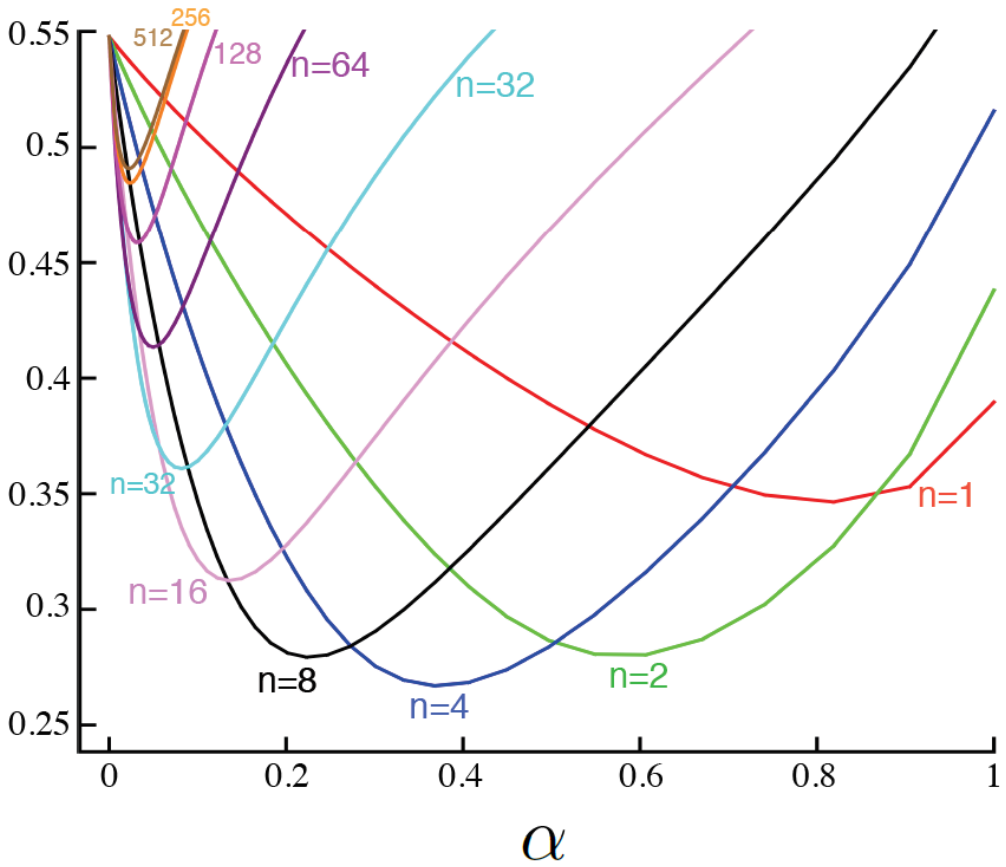
Both terminal continuation values are zero.

The plot in the next slide measures RMS error over 19 states at each episode end, then averages over **the first 10 episodes and 100 runs**.

Book Example 7.1 (Sutton & Barto, 2018, p. 145).

# n and step size interact

Average RMS error over 19 states and first 10 episodes




**Each curve fixes n. Horizontal axis: α. Lower error is better.**

**Question:** Is the best n the same at every α? What can we conclude from the low points of these curves?

Intermediate n performs best **in this experiment**. Step size matters too.

The target uses the table only if the episode is still running after n steps. A longer horizon reaches a complete return more often, but can also add sampling variability and delay.

Plot: Sutton & Barto (2018), Figure 7.2 (19-state random walk).

# Combining backup horizons

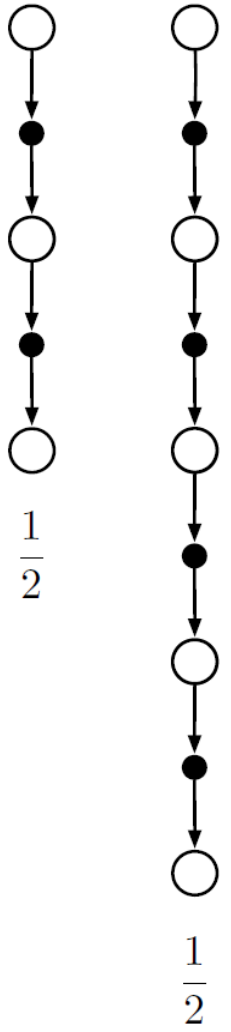
The best n depended on α. **Instead of choosing one n, can a target use several?**

A target can combine different n-step returns from the **same episode**. Example:

$$\frac{1}{2}G_{t:t+2} + \frac{1}{2}G_{t:t+4}$$

Use **nonnegative weights that sum to one**.

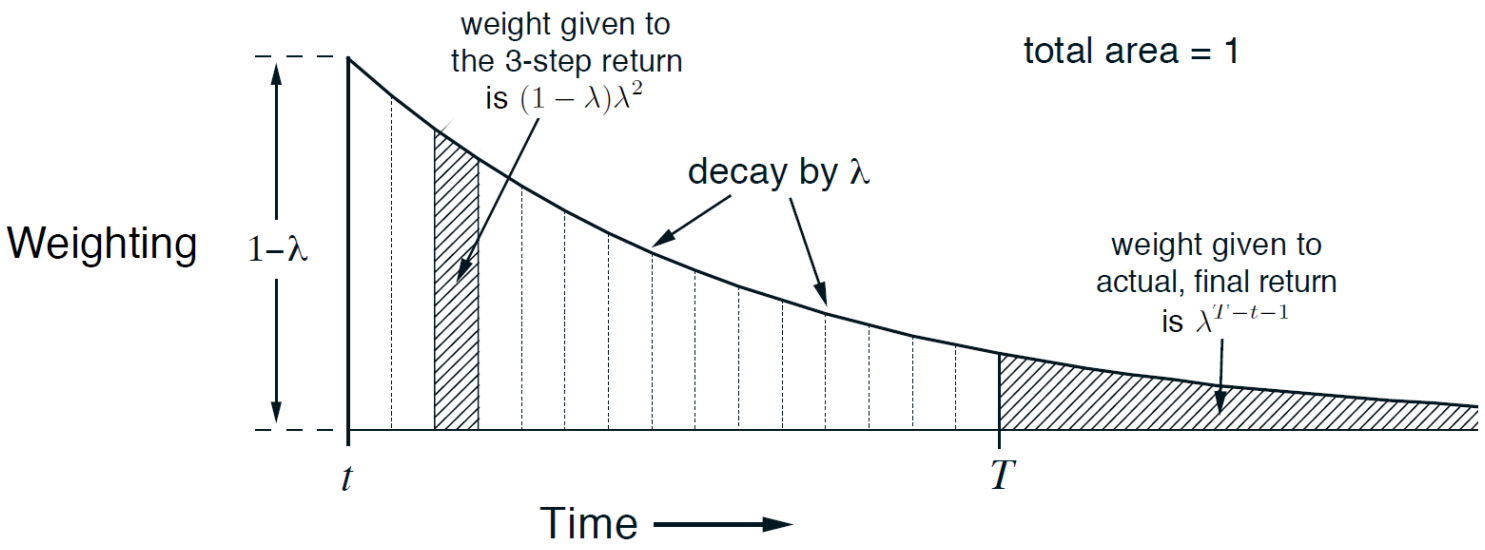
Mixing horizons does **not** guarantee a better target or remove parameter tuning.




Half two-step, half four-step

# The finite-episode λ-return

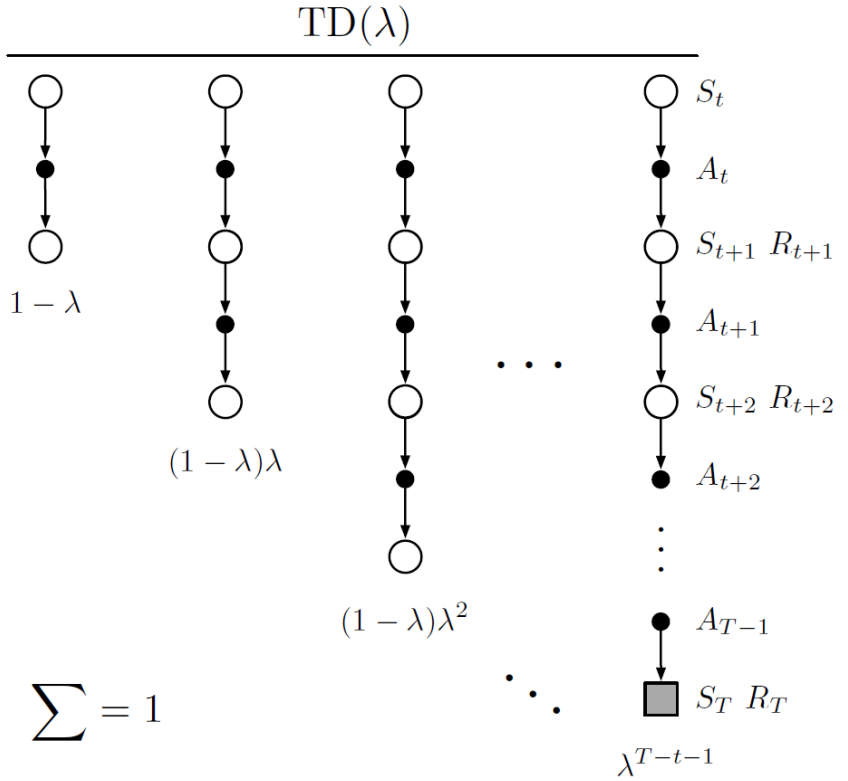
$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{T-t-1} \lambda^{n-1} G_{t:t+n} + \lambda^{T-t-1} G_t, \qquad 0 \le \lambda \le 1$$




Shorter targets: weight $(1-\lambda)\lambda^{n-1}$

Complete return: weight $\lambda^{T-t-1}$

**The final return receives all remaining weight.**




Three transitions from $t = 0$:

weights $1-\lambda$, $(1-\lambda)\lambda$, $\lambda^2$ on $G_{0:1}$, $G_{0:2}$, $G_0$.

**λ = 0:** the one-step target. **λ = 1:** the complete-return target.

Weighting diagram: Sutton & Barto (2018), Figure 12.2, p. 290. Backup diagram: Figure 12.1, p. 289.

# A λ-return on our grid

Episode (1,1) →(−0.04) (1,1) →(−0.04) (1,2) →(+1) goal, $T = 3,\ \gamma = 0.5$.

**Fixed snapshot** from MT22: V((1,1)) = −0.03, V((1,2)) = 0.5, goal 0.

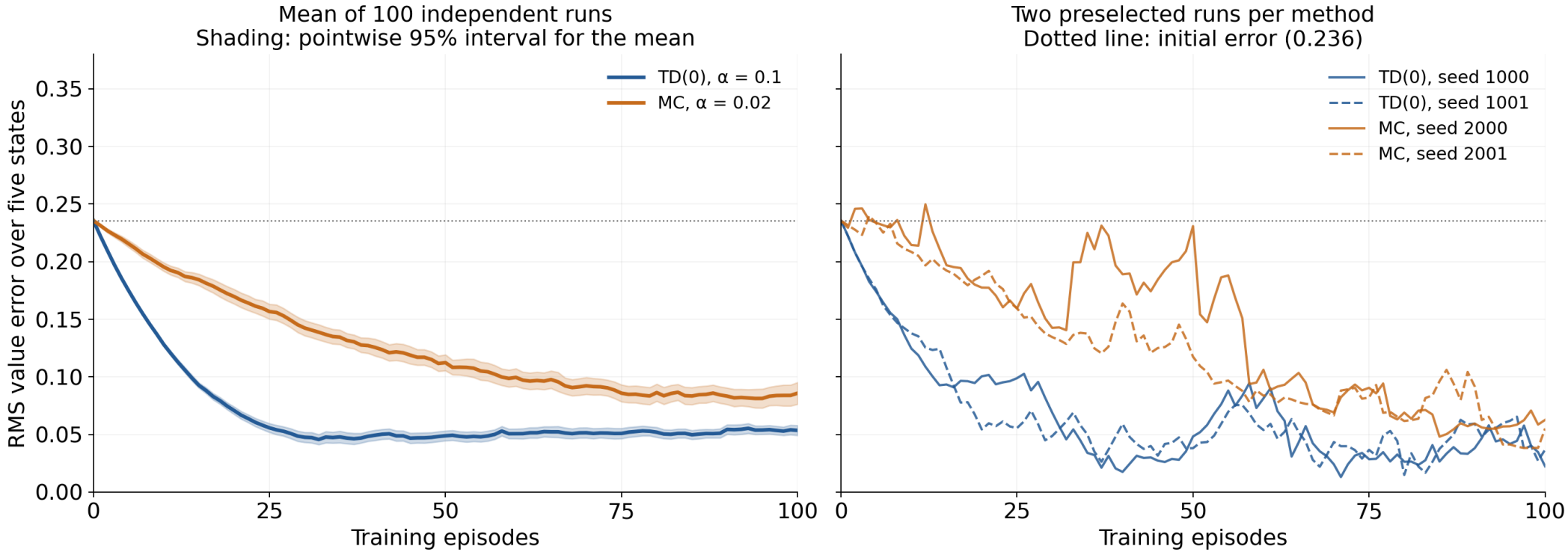
| Time-0 target | $G_{0:1}$ | $G_{0:2}$ | $G_0$ (complete) |
|---|---|---|---|
| **Value** | −0.055 | 0.065 | 0.19 |
| **Weight at λ = 0.5** | 0.5 | 0.25 | 0.25 |

**Question:** At λ = 0.5, what weight does each target get, and what is $G_0^\lambda$?

$$G_0^\lambda = 0.5(-0.055) + 0.25(0.065) + 0.25(0.19) = 0.03625$$

Check the endpoints: λ = 0 gives −0.055; λ = 1 gives 0.19.

# What this comparison supports




**Simulation of the five-state task. TD(0), α = 0.1 and every-visit MC, α = 0.02.**

"This proves TD is the better algorithm." Vote: **supported**, **unsupported**, or **needs a narrower claim**?

**Answer:** narrower. TD has lower mean error at these checkpoints for **this task, these α values, this initialization and episode budget**. MC is the baseline; the dotted line is the starting error.

# Before reporting a winner

**Question:** What would you change before reporting a winner?

Give methods **comparable tuning effort**; choose settings on development runs and report **fresh final runs**.

**State the budget:** episodes can differ in length; count transitions and distinguish computation and data passes.

# Choosing between MC, TD and n-step targets

| Question | MC | TD(0) | n-step TD |
|---|---|---|---|
| **Target** | Complete return | One reward + V | n rewards + V if still running |
| **Earliest update** | Episode end | Next transition | n transitions or episode end |
| **Target error** | Unbiased under the fixed policy | Can inherit table bias | Horizon changes reliance on the table |
| **Sampling variability** | Often high | Often lower with a useful table | Often grows with horizon; not guaranteed |
| **Useful when** | Complete episodes are available; avoid a bootstrap target | Need immediate updates or discounted continuing prediction | Can wait for more evidence before bootstrapping |
| **Dependence on V in the target** | None | Direct | Through a surviving continuation |

TD shares continuation estimates using the Markov assumption. MC's completed target avoids that bootstrap; it does not solve partial observability.

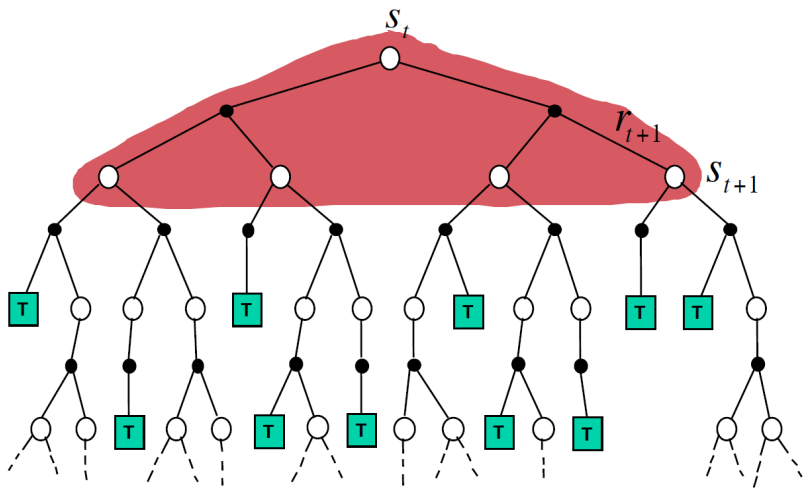
λ mixes n-step targets using geometric weights. Often helps in practice.

**Choice needs evidence:** compare runs, tuning and budgets on the task.

**Next:** control with sampled targets. Later: connect λ mixtures to advantages and GAE.

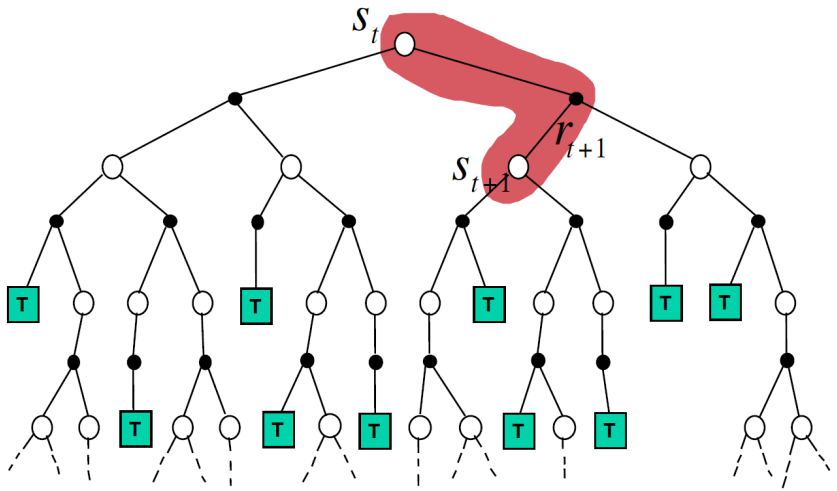
# Optional : Visual Comparison of DP, MC, TD(0) Backups

$$V(S_t) \leftarrow \mathbb{E}_\pi \left[ R_{t+1} + \gamma V(S_{t+1}) \right]$$



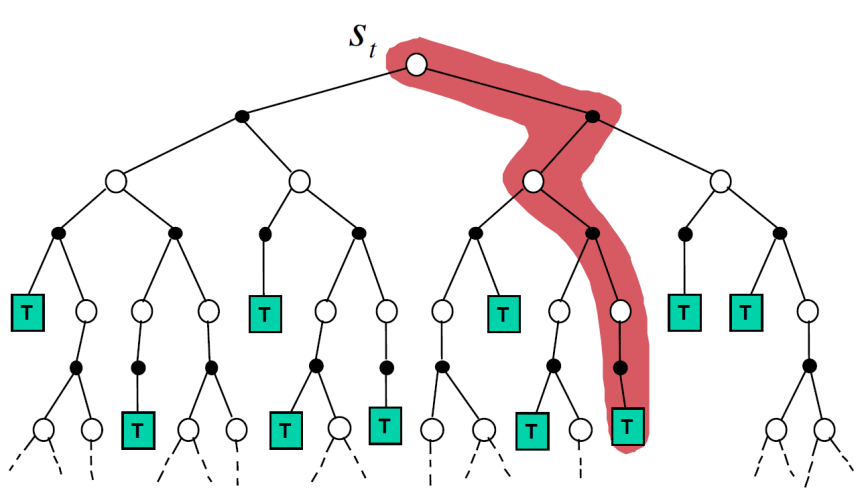

- "Breadth-wise", all potential transitions matter
- Can see why we need models and why this is computationally expensive

$$V(S_t) \leftarrow V(S_t) + \alpha \left( R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \right)$$




- "Immediate", single transition matters
- Can see why this can be better for Markovian environments

$$V(S_t) \leftarrow V(S_t) + \alpha \left( G_t - V(S_t) \right)$$

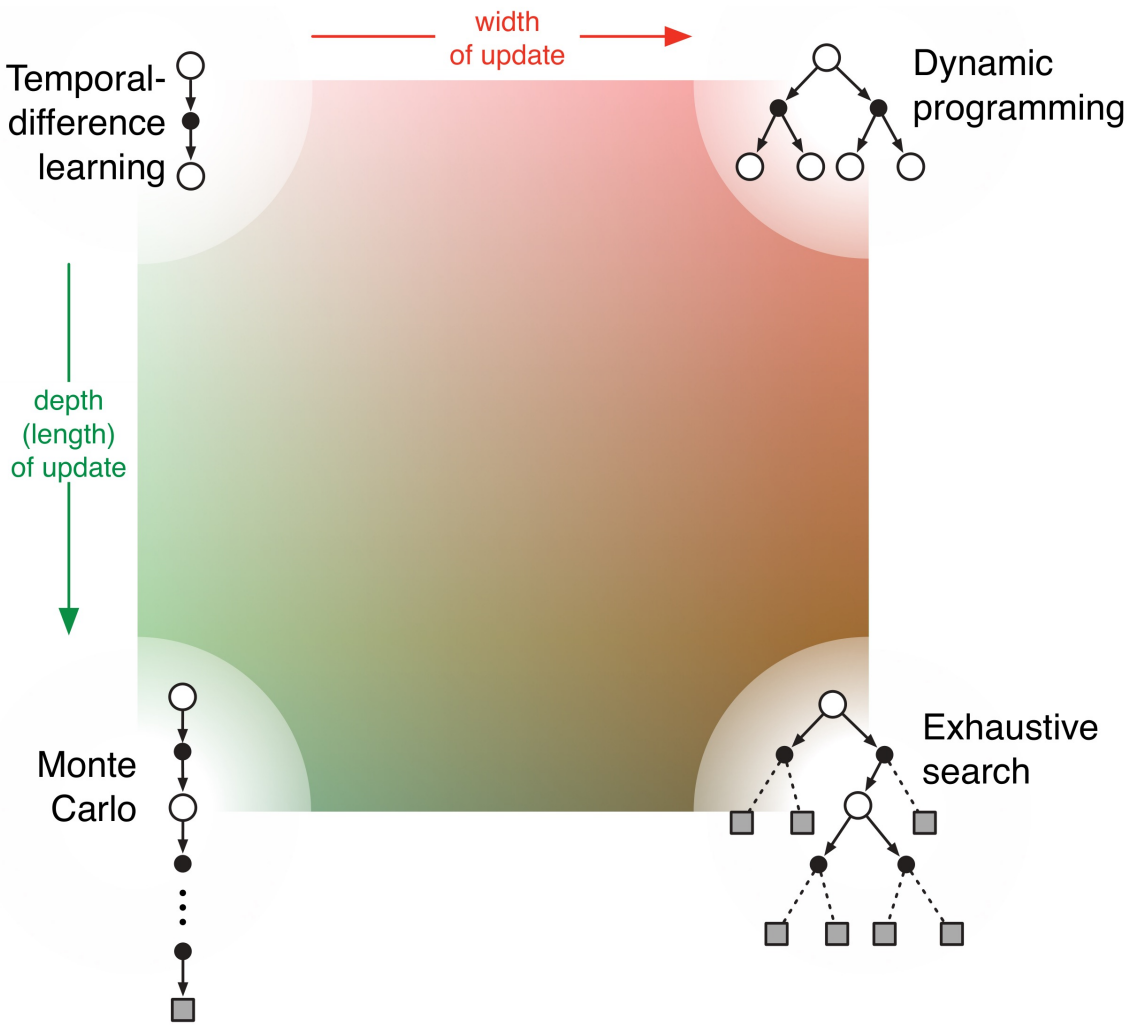



- "Depth-wise" trajectory matters
- Can see why this might be better for Non-Markovian environments

white nodes: states, black nodes: actions, T: terminal states

Figures are from David Silver's course.

# Optional: Unified View of Reinforcement Learning




- n-step TD and TD(λ) live on the left edge
- There are other methods that live somewhere in between

- What are some examples you may have seen in previous courses?

Figure: Sutton & Barto (2018), Figure 8.11, p. 190.